

Building a Stateless Firewall Using Netfilter in Linux



The Linux kernel has a Netfilter framework that allows us to perform various networking-related operations. This article is a simple tutorial on how to build firewall modules using Netfilter.

The Netfilter framework is a collection of hooks or handlers in the Linux kernel, which helps to filter or capture socket buffers. We can implement packet filtering at the input or output, or even at the forwarding path of a network packet. *Iptables* is a popular tool that is implemented using the Netfilter framework.

As shown in Figure 1, a packet can be filtered or processed at five different stages. So there are five possible hooks where programmers can attach a customised handler and implement their own firewall. These hooks are (only for Linux kernel 5.10 or above):

- ***NF_INET_PRE_ROUTING***: This hook is called once a network packet enters the stack, before any routing decision takes place.
- ***NF_INET_LOCAL_IN***: After routing, if it is found that the packet is for a local network, this hook is triggered.
- ***NF_INET_FORWARD***: This hook is called if, after routing, it is found that the packet is for another networking domain, and not for a local process.
- ***NF_INET_LOCAL_OUT***: This is called in case the packet is sent from a local process using *send* or *sendto* (POSIX calls).
- ***NF_INET_POST_ROUTING***: This handler is called just before any local or forwarded packet is about to hit the interface after handling by the entire stack is over.

I have written an example code to show how to build a firewall using the Netfilter framework. I have used Linux kernel 5.10. In this example I have blocked all ICMP and HTTP/HTTPS packet sending from a local process. This program has to run from kernel space, and not in user space. So a kernel module has been developed.

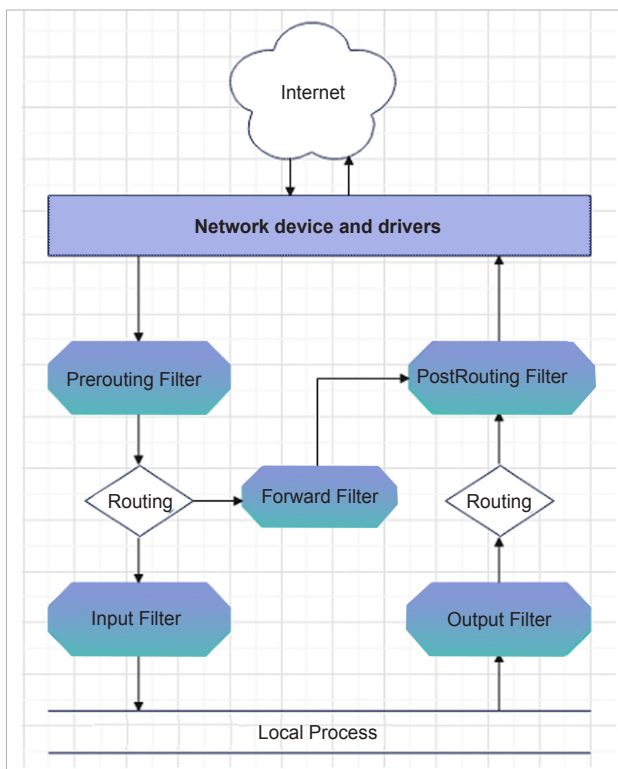


Figure 1: Processing stages